



**SYDNEY BOYS HIGH SCHOOL**  
**MOORE PARK, SURRY HILLS**

**2014**  
**HIGHER SCHOOL CERTIFICATE**  
**TRIAL PAPER**

# Mathematics      Extension 2

## General Instructions

- Reading Time – 5 Minutes
- Working time – 3 Hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Answer Questions 1 to 10 on the sheet provided.
- Each Question from 11 to 16 is to be returned in a separate bundle.
- All necessary working should be shown in every question

## Total Marks – 100

- Attempt questions 1 – 16
- Answer in simplest exact form unless otherwise instructed

Examiner: *P.R.Bigelow*

- NOTE: This is a trial paper only and does not necessarily reflect the content or format of the final Higher School Certificate examination paper for this subject.

## Use Multiple Choice Answer Sheet

### Question 1

Seven people are to be placed in four hotel rooms.  
In how many ways may this be done?

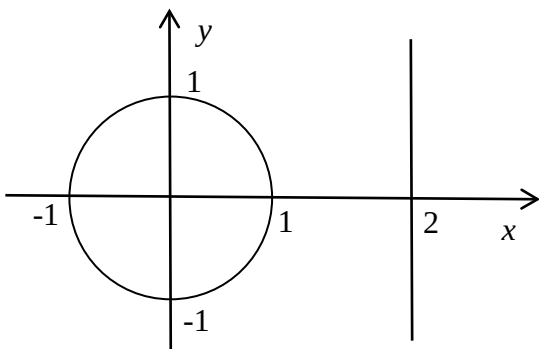
- A:  $4^7$
- B:  ${}^7C_4$
- C:  ${}^7P_4$
- D:  $7^4$

### Question 2

$$i^{2114} =$$

- A: 1
- B:  $i$
- C:  $-i$
- D:  $-1$

### Question 3



The circle  $x^2 + y^2 = 1$  is rotated about the line  $x = 2$ . With use of cylindrical shells, the volume is given by:

- A:  $4\pi \int_{-1}^1 (2-x)\sqrt{1-x^2} dx$
- B:  $8\pi \int_0^1 (2-x)\sqrt{1-x^2} dx$
- C:  $2\pi \int_{-1}^1 (2-x)\sqrt{1-x^2} dx$
- D:  $4\pi \int_1^2 (2-x)\sqrt{1-x^2} dx$

#### Question 4

The equation of the chord of contact from  $(5, -2)$  to the ellipse  $\frac{x^2}{16} + \frac{y^2}{9} = 1$  is given by:

A:  $\frac{x}{8} - \frac{5y}{16} = 1$

B:  $\frac{5x}{16} + \frac{2y}{9} = 1$

C:  $\frac{5x}{16} - \frac{2y}{9} = 0$

D:  $\frac{5x}{16} - \frac{2y}{9} = 1$

#### Question 5

The roots of  $x^3 + 5x + 11 = 0$  are  $\alpha, \beta$ , and  $\gamma$ .

The value of  $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$  is:

A: 25

B: 0

C: -55

D: 55

#### Question 6

If  $a$  and  $b$  are positive, which of the following is false?

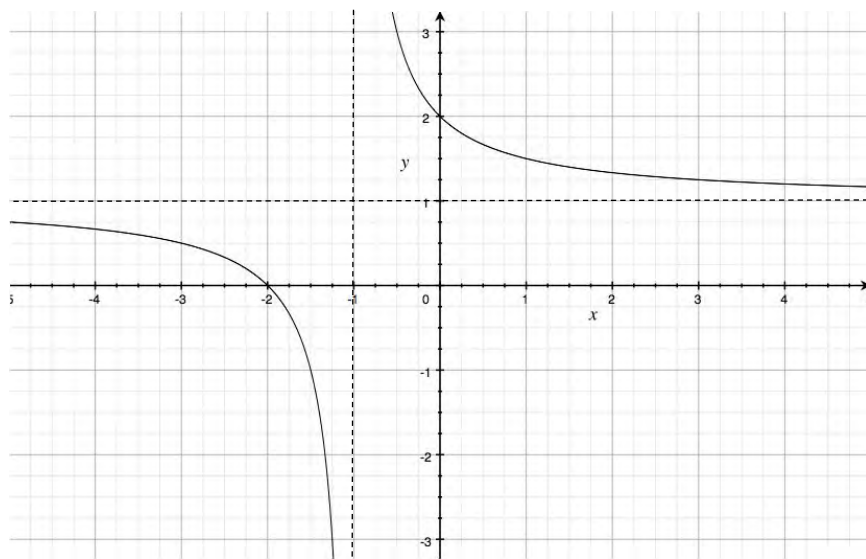
A:  $\frac{a}{b} + \frac{b}{a} \geq 2$ .

B:  $\frac{a+b}{2} \leq \sqrt{ab}$ .

C:  $(\sqrt{a} - \sqrt{b})^2 \geq 2ab$ .

D:  $(a+b)^2 \geq (a-b)^2 + (2ab)^2$ .

### Question 7



The graph has equation:

A:  $(x-1)(y+1)=1$

B:  $y = \frac{x+2}{x}$

C:  $(x+1)(y-1)=1$

D:  $y = \frac{x}{x+1}$

### Question 8

$1+i$  is a zero of  $x^3+ax+b$  where  $a, b$  are real, therefore the values of  $a$  and  $b$  are:

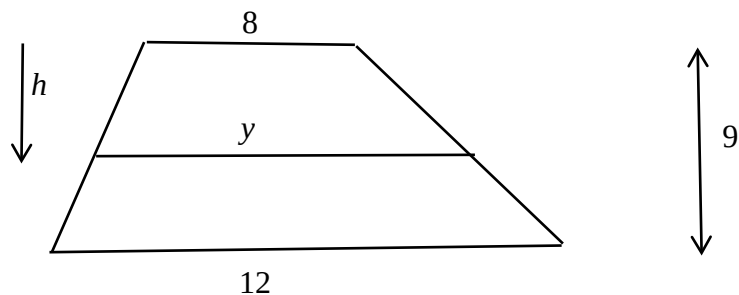
A:  $a=-2, b=-4$

B:  $a=-2, b=4$

C:  $a=2, b=-4$

D:  $a=2, b=4$

### Question 9



The diagram shows a trapezium, with an internal parallel line. Which of the following is true?

A:  $y = \frac{3}{4}h + 8.$

B:  $y = \frac{3}{4}h + 9.$

C:  $4y = 9h + 72$

D:  $9y = 4h + 72$

### Question 10

By considering the graphs of  $y = 3x^2 - 2x - 2$  and  $y = |3x|$ , the solution to  $3x^2 - 2x - 2 \leq |3x|$  is:

A:  $-\frac{1}{3} \leq x \leq 2.$

B:  $-1 \leq x \leq \frac{3}{2}.$

C:  $-\frac{1}{3} \leq x \leq \frac{3}{2}$

D:  $-1 \leq x \leq 2$

**Question 11.** (15 marks) (Start a new answer booklet.)

**Marks**

- (a) Given  $z = 1 - i$ , find the values of  $w$  such that

**2**

$$w^2 = i + 3\bar{z}$$

- (b) On separate Argand diagrams, shade the following regions:

(i)  $4 \leq z + \bar{z} \leq 10$

**1**

(ii)  $\arg(z^2) = \frac{2\pi}{3}$

**1**

(iii)  $z\bar{z} = 4$

**1**

- (c) (i) Show that  $z = 1 + i$  is a root of the polynomial

$$z^2 - (3 - 2i)z + (5 - i) = 0$$

**1**

- (ii) Find the other root.

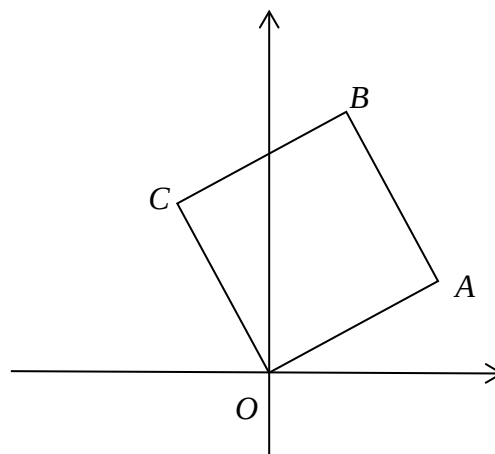
**1**

- (d)  $OABC$  is a square in the Argand diagram.

**3**

$B$  represents the complex number  $2 + 2i$ .

Find the complex numbers represented by  $A$  and  $C$ .



### Question 11 (Continued)

(e) From  $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$  codes of three digits are formed, where no digit is repeated.

(i) Find the number of possible different codes. 1

(ii) How many of these are *not* in decreasing order of magnitude, reading from left to right? 2

(f) Given that  $\alpha, \beta$ , and  $\gamma$  are the roots of  $x^3 - 7x + 6 = 0$ , evaluate

$$\alpha^3 + \beta^3 + \gamma^3 \quad \quad \quad 2$$

**Question 12.** (15 marks) (Start a new answer booklet.)

**Marks**

- (a) Find  $\int x e^{4x} dx$ . 2
- (b) Evaluate  $\int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \frac{dx}{\cos x + 2}$ . 2
- (c) Find  $\int \frac{du}{8 + u^3}$ . 2
- (d) Evaluate  $\int_0^{\frac{\pi}{4}} \cos^5 \theta d\theta$  2
- (e) (i) Find  $\int \frac{dx}{x^2 + 2x + 10}$ . 1
- (ii) Hence find  $\int \frac{x^2}{x^2 + 2x + 10} dx$ . 2
- (f) Consider the curve defined by  $2x^2 + xy - y^2 = 0$ . 2
- Find the values of  $\frac{dy}{dx}$  and  $\frac{d^2y}{dx^2}$  at the point  $P(2, 4)$ .
- (g) Sketch the locus  $|z - 1| + |z + 1| = 4$  (where  $z$  is a complex number), showing  $x$  and  $y$  intercepts. 2

**Question 13.** (15 marks) (Start a new answer booklet.)

**Marks**

- (a) Find the values of the real numbers  $p$  and  $q$  given that

**2**

$$x^3 + 2x^2 - 15x - 36 = (x + p)^2(x + q)$$

- (b) An ellipse has equation

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

- (i) Find the eccentricity of the ellipse.

**1**

- (ii) Sketch the ellipse showing foci, directrices and intercepts.

**2**

- (iii) Prove that the equation of the tangent to the ellipse at the point  $P(3\cos\theta, 2\sin\theta)$  is  $2x\cos\theta + 3y\sin\theta = 6$ .

**3**

- (iv) The ellipse meets the  $y$ -axis at points  $A$  and  $B$ . The tangents to the ellipse at  $A$  and  $B$  meet the tangent at  $P$ , at points  $C$  and  $D$  respectively.

**3**

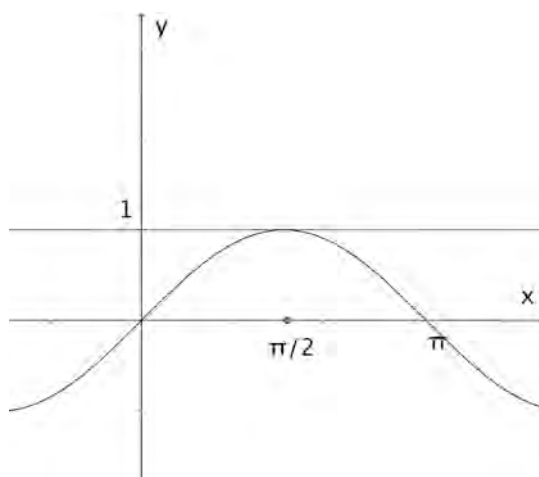
Prove that  $AC \times BD = 9$ .

- (c) The area defined by  $0 \leq x \leq \frac{\pi}{2}$ ,  $0 \leq y \leq 1$  and  $y \geq \sin x$  is rotated about the line  $y = 1$ .

**4**

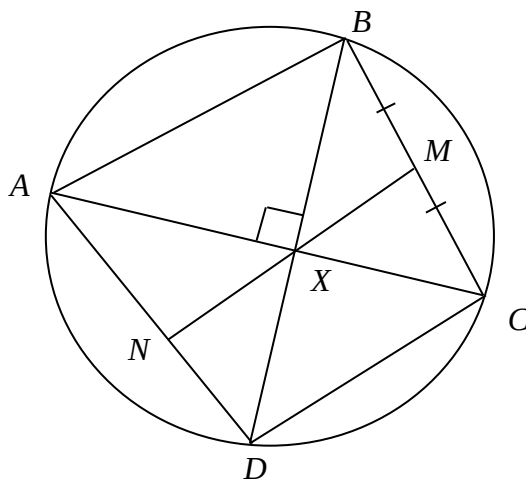
- (i) Copy the diagram and shade the defined area.

- (ii) Find the volume of the solid by taking slices perpendicular to the axis of rotation.



**Question 14** (15 marks) (Start a new answer booklet.)

- (a)  $ABCD$  is a cyclic quadrilateral. The diagonals  $AC$  and  $BD$  intersect at right-angles at  $X$ .  $M$  is the mid-point of  $BC$ , and  $MX$  produced meets  $AD$  at  $N$ .



- (i) Copy the diagram to your answer booklet, then show that  $BM = MX$ . 1
- (ii) Show that  $\angle MBX = \angle MXB$ . 1
- (iii) Show that  $MN$  is perpendicular to  $AD$ . 3
- (b) The base of a solid is in the shape of an ellipse with equation  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ .  
Vertical cross-sections taken perpendicular to the major axis are rectangles where length is double the height.
- (i) Show that the volume of a typical rectangular slice is 2
- $$\delta V = \frac{2b^2}{a^2}(a^2 - x^2)\delta x$$
- (where  $\delta x$  is the width of the slice.)
- (ii) Find the volume of the solid by integration. 2

### Question 14 (Continued)

(c) In each of the following parts,  $x, y, z, w, a, b, c, d > 0$ :

(i) Show that  $(x + y)^2 \geq 4xy$ . 1

(ii) Show that  $[(x + y)(z + w)]^2 \geq 16xyzw$  1

(iii) Deduce that  $\frac{x + y + z + w}{4} \geq \sqrt[4]{xyzw}$  2

(iv) Hence show that (using (iii)): 2

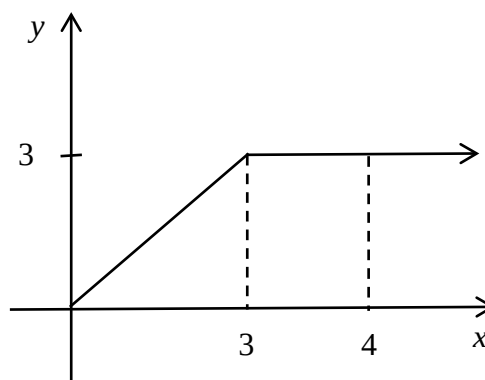
$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4$$

**Question 15 (15 marks) (Start a new answer booklet.)**

**Marks**

- (a) The graph of  $y = f(x)$  is shown.

Sketch the following on separate diagrams:



(i)  $y = f(4 - x)$ .

**1**

(ii)  $y = f(|x|)$ .

**1**

(iii)  $y \times f(x) = 1$ .

**1**

(iv)  $y^2 = f(x)$ .

**1**

- (b) Let  $w$  be a non-real cube root of unity.

(i) Show that  $1 + w + w^2 = 0$ .

**1**

(ii) Simplify  $(1 + w)^2$ .

**1**

(iii) Show that  $(1 + w)^3 = -1$ .

**1**

(iv) Using part (iii) simplify  $(1 + w)^{3n}$  where  $n \in \mathbb{Z}^+$ .

**1**

(v) Show that

**3**

$$\binom{3n}{0} - \frac{1}{2} \left[ \binom{3n}{1} + \binom{3n}{2} \right] + \binom{3n}{3} - \frac{1}{2} \left[ \binom{3n}{4} + \binom{3n}{5} \right] + \binom{3n}{6} - \dots$$

$$\dots + \binom{3n}{3n} = (-1)^n$$

[Hint: You may use  $\operatorname{Re}(w) = \operatorname{Re}(w^2) = -\frac{1}{2}$ ]

- (c) (i) Show that  $\ln(ex) > e^{-x}$  for  $x \geq 1$ . (Use a diagram.)

**1**

(ii) Hence show that  $\ln(e^n \times n!) > \frac{e^n - 1}{e^n(e - 1)}$ .

**3**

**Question 16 (15 marks) (Start a new answer booklet.)**

- (a) A Particle  $P$  of unit mass is thrown vertically downwards in a medium where the resistive force is proportional to the velocity.

(i) Taking *downwards as positive*, show that  $\ddot{x} = g - kv$  for some  $k > 0$ . 1

(ii) Given that the initial speed is  $U$  and the particle is thrown from a point  $T$ , distant  $d$  units above a fixed point  $O$ , (taken as the Origin) so that the initial conditions are  $v = U$  and  $x = -d$ . 2

$$\text{Show that } v = \frac{g}{k} - \left( \frac{g - kU}{k} \right) e^{-kt}.$$

(iii) Hence show that: 2

$$x = \frac{gt - kd}{k} + \left( \frac{g - kU}{k^2} \right) (e^{-kt} - 1)$$

(iv) A second identical particle  $Q$  is dropped from  $O$ , at then same instant that  $P$  is thrown down. Use the above results to find expressions for  $v$  and  $x$  as functions of  $t$ , for the particle  $Q$ . 2

(v) The particles collide. Find when this occurs, and find the speed at which they collide 3

(b) (i) Show that: 1

$$\sin(2r+1)\theta - \sin(2r-1)\theta = 2\sin\theta\cos 2r\theta$$

(ii) Hence shown that: 2

$$\sin\theta \sum_{r=1}^n \cos 2r\theta = \frac{1}{2} \{ \sin(2n+1)\theta - \sin\theta \}.$$

(iii) Hence evaluate: 2

$$\sum_{r=1}^{100} \cos^2 \frac{r\pi}{100}.$$

**This is the end of the paper.**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right) \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x, \quad x > 0$



Student Number: \_\_\_\_\_ **SOLUTIONS**

## Mathematics Extension 2 Trial HSC 2014

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

**Sample:**  $2 + 4 =$  (A) 2 (B) 6 (C) 8 (D) 9  
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐  
correct

### Section I: Multiple choice answer sheet.

Completely colour the cell representing your answer. Use black pen.

1. ☒ (B) ☐ (C) ☐ (D)
2. ☐ (A) ☐ (B) ☐ (C) ☒
3. ☒ (B) ☐ (C) ☐ (D)
4. ☐ (A) ☐ (B) ☐ (C) ☒
5. ☐ (A) ☒ (C) ☐ (D)
6. ☒ (B) ☐ (C) ☐ (D)
7. ☐ (A) ☐ (B) ☒ (D)
8. ☐ (A) ☒ (C) ☐ (D)
9. ☐ (A) ☐ (B) ☐ (C) ☒
10. ☐ (A) ☐ (B) ☐ (C) ☒

2014 Extension 2 Mathematics Trial HSC:  
Solutions— Question 11

11. (a) Given  $z = 1 - i$ , find the values of  $w$  such that

2

$$w^2 = i + 3\bar{z}.$$

**Solution:**  $i + 3\bar{z} = i + 3 + 3i,$   
 $= 3 + 4i,$   
 $= w^2.$

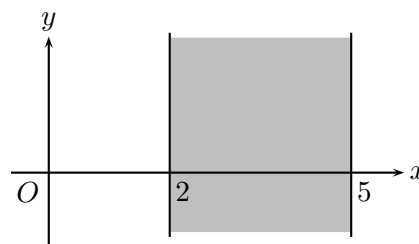
Let  $w = a + ib;$   
 $a^2 - b^2 + 2abi = 3 + 4i,$   
 $a^2 - b^2 = 3,$   
 $a^2 + b^2 = 5,$   
 $ab = 2,$   
 $2a^2 = 8,$   
 $a = \pm 2,$   
 $b = \pm 1.$   
 $\therefore w = \pm(2 + i).$

- (b) On separate Argand diagrams, shade the following regions:

- (i)  $4 \leq z + \bar{z} \leq 10$

1

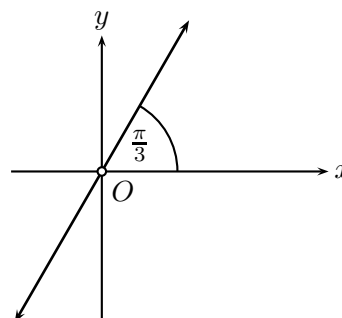
**Solution:**  $z + \bar{z} = 2\Re(z),$   
 $4 \leq 2\Re(z) \leq 10,$   
 $2 \leq \Re(z) \leq 5.$



- (ii)  $\arg(z^2) = \frac{2\pi}{3}$

1

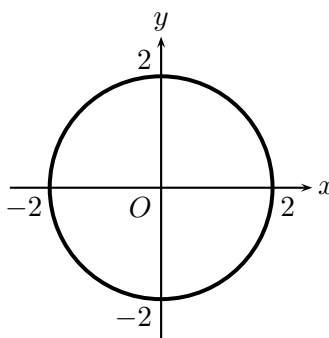
**Solution:**  $\arg(z^2) = 2\arg(z),$   
 $= \frac{2\pi}{3}, -\frac{4\pi}{3},$   
 $\therefore \arg(z) = \frac{\pi}{3}, -\frac{2\pi}{3}.$



(iii)  $z\bar{z} = 4$

1

**Solution:**  $z\bar{z} = |z|^2,$   
 $|z| = 2.$



(c) (i) Show that  $z = 1 + i$  is a root of the polynomial

1

$$z^2 - (3 - 2i)z + (5 - i) = 0.$$

**Solution:** Put  $P(z) = z^2 - (3 - 2i)z + (5 - i),$   
 $P(1 + i) = (1 + i)^2 - (3 - 2i)(1 + i) + 5 - i,$   
 $= 1 + 2i - 1 - (3 + 3i - 2i + 2) + 5 - i,$   
 $= i + 5 - 5 - i,$   
 $= 0.$   
*i.e.*  $1 + i$  is a root of  $P(z).$

(ii) Find the other root.

1

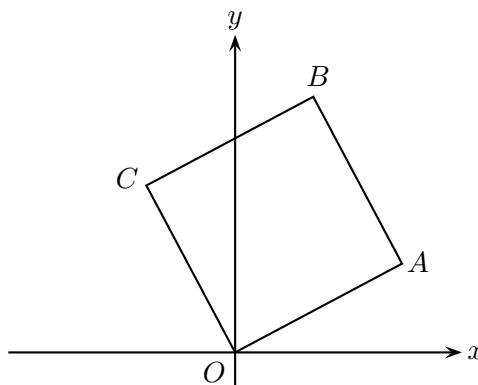
**Solution:** Let the other root be  $a + ib.$   
 $1 + i + a + ib = 3 - 2i,$   
 $1 + a = 3,$   
 $1 + b = -2,$   
 $a = 2,$   
 $b = -3.$   
 $\therefore$  The other root is  $2 - 3i.$

(d)  $OABC$  is a square in the Argand diagram.

3

$B$  represents the complex number  $2 + 2i.$

Find the complex numbers represented by  $A$  and  $C.$



**Solution: Method 1—**

Notice that  $\arg(B) = \frac{\pi}{4}$ , so that  $A$  must lie on  $Ox$  and  $C$  must lie on  $Oy$ . Hence  $A = (2 + 0i)$  and  $C = (0 + 2i).$

**Solution: Method 2—**

Let  $A = z = a + ib$  and so  $C = iz = -b + ai$ .

$$B = A + C,$$

$$2 + 2i = a - b + i(a + b),$$

$$a - b = 2,$$

$$a + b = 2,$$

$$2a = 4,$$

$$a = 2,$$

$$b = 0.$$

*i.e.*  $A = (2 + 0i)$  and  $C = (0 + 2i)$ .

**Solution: Method 3—**

$$|B| = \sqrt{2^2 + 2^2},$$

$$= 2\sqrt{2}.$$

$$|A| = 2.$$

$$A = \frac{B}{\sqrt{2}} \operatorname{cis} \left( -\frac{\pi}{4} \right),$$

$$= \frac{2 + 2i}{\sqrt{2}} \left( \frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}} \right),$$

$$= \frac{2 - 2i + 2i + 2}{\sqrt{2} \times \sqrt{2}},$$

$$= 2 + 0i.$$

$$C = Ai,$$

$$= 0 + 2i.$$

- (e) From  $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$  codes of three digits are formed, where no digit is repeated.

- (i) Find the number of possible different codes.

1

**Solution:**  $9 \times 8 \times 7 = 504$  different codes.

- (ii) How many of these are *not* in decreasing order of magnitude, reading from left to right?

2

**Solution:** 6 ways of arranging any group of 3, only one of which is in decreasing order of magnitude.

There are  ${}^9C_3 = 84$  ways of selecting groups of 3.

Thus there are  $5 \times 84 = 420$  which are not decreasing.

(f) Given that  $\alpha$ ,  $\beta$ , and  $\gamma$  are the roots of  $x^3 - 7x + 6 = 0$ , evaluate

2

$$\alpha^3 + \beta^3 + \gamma^3.$$

**Solution: Method 1—**

$$\begin{aligned}\alpha + \beta + \gamma &= 0, \\ \alpha\beta + \beta\gamma + \gamma\alpha &= -7, \\ \alpha\beta\gamma &= -6.\end{aligned}$$

As  $\alpha$ ,  $\beta$ , and  $\gamma$  are roots,

$$\begin{aligned}\alpha^3 - 7\alpha + 6 &= 0, \\ \beta^3 - 7\beta + 6 &= 0, \\ \gamma^3 - 7\gamma + 6 &= 0, \\ \alpha^3 + \beta^3 + \gamma^3 - 7(\alpha + \beta + \gamma) + 18 &= 0, \\ \alpha^3 + \beta^3 + \gamma^3 &= -18.\end{aligned}$$

**Solution: Method 2—**

$$\begin{aligned}\text{Put } y &= x^3, \\ x &= y^{\frac{1}{3}}. \\ y - 7y^{\frac{1}{3}} + 6 &= 0, \\ y + 6 &= 7y^{\frac{1}{3}}, \\ y^3 + 18y^2 + 108y + 216 &= 343y, \\ y^3 + 18y^2 - 235y + 216 &= 0. \\ \therefore \alpha^3 + \beta^3 + \gamma^3 &= -18.\end{aligned}$$

**Solution: Method 3—**

$$\begin{aligned}(\alpha + \beta + \gamma)^3 &= (\alpha^2 + \beta^2 + \gamma^2 + 2\alpha\beta + 2\alpha\gamma + 2\beta\gamma)(\alpha + \beta + \gamma), \\ &= \alpha^3 + \alpha\beta^2 + \alpha\gamma^2 + 2\alpha^2\beta + 2\alpha^2\gamma + 2\alpha\beta\gamma + \alpha^2\beta + \\ &\quad \beta^3 + \beta\gamma^2 + 2\alpha\beta^2 + 2\alpha\beta\gamma + 2\beta^2\gamma + \alpha^2\beta + \beta^2\gamma + \\ &\quad \gamma^3 + 2\alpha\beta\gamma + 2\alpha\gamma^2 + 2\beta\gamma^2, \\ &= \alpha^3 + \beta^3 + \gamma^3 + 6\alpha\beta\gamma + 3(\alpha\beta^2 + \alpha\gamma^2 + \alpha^2\beta + \beta\gamma^2 + \\ &\quad \alpha^2\gamma + \beta^2\gamma), \\ &= \alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma + 3(\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma), \\ \alpha^3 + \beta^3 + \gamma^3 &= 3\alpha\beta\gamma + (\alpha + \beta + \gamma)^3 - 3(\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma), \\ &= 3(-6) + 0^3 - 3(0)(-7), \\ &= -18.\end{aligned}$$

QUESTION TWELVE.

$$\int x e^{4x} dx$$

$$\text{let } u = x \quad dv = e^{4x} dx$$

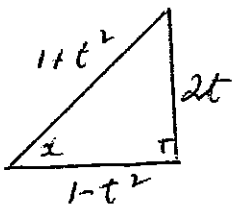
$$du = dx \quad v = \frac{1}{4} e^{4x}$$

$$\begin{aligned} \int &= \frac{1}{4} x e^{4x} - \frac{1}{4} \int e^{4x} dx \\ &= \frac{1}{4} x e^{4x} - \frac{1}{16} e^{4x} + C \end{aligned}$$

$$2) \int_{\frac{3\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{\cos x + 2}$$

$$\text{let } t = \tan \frac{x}{2}$$

$$\tan x = \frac{2t}{1-t^2}$$



$$\cos x = \frac{1-t^2}{1+t^2} \text{ from diagram}$$

$$\begin{aligned} \frac{dt}{dx} &= \frac{1}{2} \sec^2 \frac{x}{2} \\ &= \frac{1}{2} (1 + \tan^2 \frac{x}{2}) \\ &= \frac{1}{2} (1 + t^2) \end{aligned}$$

$$\frac{2dt}{1+t^2} = dx$$

$$\text{When } x = \frac{5\pi}{2}, t = 1$$

$$x = \frac{3\pi}{2}, t = -1$$

$$\int = 2 \int_{-1}^1 \frac{dt}{(1+t^2) \left[ \frac{1-t^2}{1+t^2} + 2 \right]}$$

$$= 2 \int_{-1}^1 \frac{dt}{1-t^2+2+2t^2}$$

$$= 2 \int_{-1}^1 \frac{dt}{t^2+3}$$

$$= \frac{2}{\sqrt{3}} \left[ \tan^{-1} \frac{t}{\sqrt{3}} \right]_{-1}^1$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{1}{\sqrt{3}} \right) + \frac{2}{\sqrt{3}} \tan^{-1} \left( \frac{1}{\sqrt{3}} \right)$$

$$= \frac{4}{\sqrt{3}} \cdot \frac{\pi}{6}$$

$$= \frac{2\pi}{3\sqrt{3}} = \frac{2\sqrt{3}\pi}{9}$$

QUESTION 12 c

Consider  $\frac{1}{u^3+8} = \frac{A}{u+2} + \frac{Bu+C}{u^2-2u+4}$

$$= \frac{Au^2-2Au+4A+Bu^2+Cu+2Bu+2C}{u^3+8}$$

$$= \frac{u^2(A+B) + u(C+2B-2A) + 4A+2C}{u^3+8}$$

$$A+B=0$$

$$2A-2B-C=0$$

$$4A+2C=1$$

Solving  $\Rightarrow A=\frac{1}{12}, B=-\frac{1}{12}, C=\frac{1}{3}$

Then  $\frac{1}{u^3+8} = \frac{\frac{1}{12}}{u+2} + \frac{-\frac{1}{12}u+\frac{1}{3}}{u^2-2u+4}$

$$= \frac{\frac{1}{12}}{u+2} - \frac{1}{12} \left( \frac{u-4}{u^2-2u+4} \right)$$

$$= \frac{\frac{1}{12}}{u+2} - \frac{1}{12} \left[ \frac{\frac{1}{2} \left( \frac{2u-2}{u^2-2u+4} \right) - \frac{3}{u^2-2u+4}}{\right]$$

$$= \frac{\frac{1}{12}}{u+2} - \frac{1}{24} \left( \frac{2u-2}{u^2-2u+4} \right) + \frac{1}{4} \left( \frac{1}{(u-1)^2+3} \right)$$

Then

$$\int \frac{du}{u^3+8} = \frac{1}{12} \ln(u+2) - \frac{1}{24} \ln(u^2-2u+4) + \frac{1}{4\sqrt{3}} \tan^{-1} \frac{(u-1)}{\sqrt{3}}$$

TWELVE d,

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \cos^5 \theta \, d\theta &= \int \cos^2 \theta \cdot \cos^2 \theta \cos \theta \, d\theta \\&= \int (1 - \sin^2 \theta)^2 \cos \theta \, d\theta \\&= \int (1 - 2\sin^2 \theta + \sin^4 \theta) \cos \theta \, d\theta \\&= \sin \theta - \frac{2}{3} \sin^3 \theta + \frac{1}{5} \sin^5 \theta \Bigg|_0^{\frac{\pi}{4}} \\&= \frac{1}{\sqrt{2}} - \frac{2}{3} \left( \frac{1}{\sqrt{2}} \right)^3 + \frac{1}{5} \left( \frac{1}{\sqrt{2}} \right)^5 \\&= \frac{1}{\sqrt{2}} - \frac{2}{3} \times \frac{1}{2\sqrt{2}} + \frac{1}{5} \times \frac{1}{4\sqrt{2}} \\&= \frac{1}{\sqrt{2}} - \frac{1}{3\sqrt{2}} + \frac{1}{20\sqrt{2}} \\&= \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{6} + \frac{\sqrt{2}}{40} \\&= \cancel{\frac{20\sqrt{2}}{40}} \\&= \frac{60\sqrt{2}}{120} - \frac{20\sqrt{2}}{120} + \frac{3\sqrt{2}}{120} \\&= \frac{43\sqrt{2}}{120}\end{aligned}$$

TWELVE e

$$\begin{aligned}\sqrt{\frac{dx}{x^2+2x+10}} &= \sqrt{\frac{dx}{(x+1)^2+9}} \\ &= \frac{1}{3} \tan^{-1} \frac{x+1}{3} + C\end{aligned}$$

AND

$$\begin{aligned}\frac{x^2}{x^2+2x+10} &= \frac{x^2+2x+10}{x^2+2x+10} - \frac{2x+10}{x^2+2x+10} \\ &= 1 - \frac{2x+2}{x^2+2x+10} - \frac{8}{x^2+2x+10} \\ &= 1 - \frac{2x+2}{x^2+2x+10} - \frac{8}{(x+1)^2+9}\end{aligned}$$

AND

$$\begin{aligned}\sqrt{\frac{x^2 dx}{x^2+2x+10}} &= x - \ln(x^2+2x+10) - \frac{8}{3} \tan^{-1} \frac{x+1}{3} \\ &\quad + C\end{aligned}$$

QUESTION 12 f

$$2x^2 + xy - y^2 = 0$$

$$4x + x \frac{dy}{dx} + y - 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (2y - x) = 4x + y$$

$$\frac{dy}{dx} = \frac{4x + y}{2y - x}$$

$$= \frac{8+4}{8-2} \text{ at } P(2,4)$$

$$= 2$$

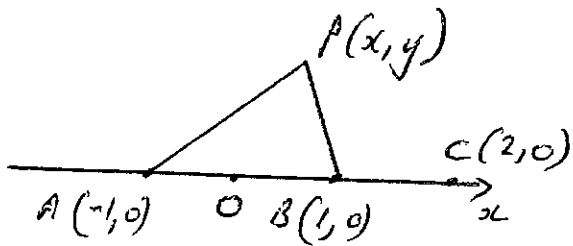
$$\frac{d^2y}{dx^2} = \frac{(2y-x) \left(4 + \frac{dy}{dx}\right) - (4x+y) \left(2 \frac{dy}{dx} - 1\right)}{(2y-x)^2}$$

$$= \frac{(6)(6) - (12)(3)}{36}$$

$$= 0$$

QUESTION TWELVE j.

$$|3-1| + |3+1| = 4$$

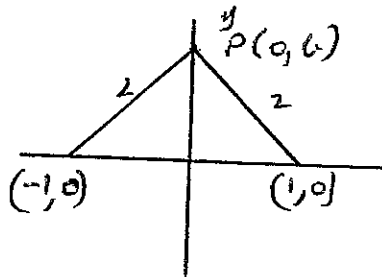


$$PA + PB = 4$$

P has position on the x axis

where  $y=0$  of  $C(2,0)$

Hence major axis has length 4



Let P have position  $(0, b)$  on the y axis

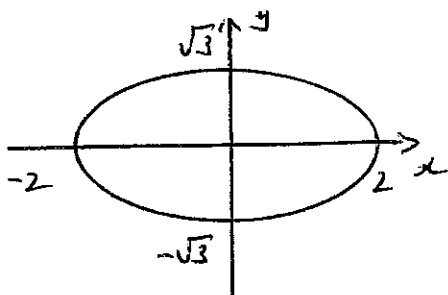
Then

$$\sqrt{b^2+1} + \sqrt{b^2+1} = 4$$

$$\sqrt{b^2+1} = 2$$

$$b^2+1 = 4$$

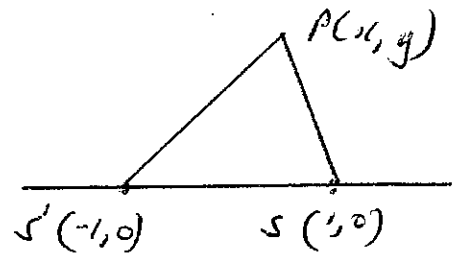
$$b = \sqrt{3}$$



hence is ellipse

$$\frac{x^2}{4} + \frac{y^2}{3} = 1$$

ALTERNATIVELY  
USING DISTANCES



$$PS' + PS = 4$$

$$\sqrt{(x-1)^2+y^2} + \sqrt{(x+1)^2+y^2} = 4$$

$$\sqrt{(x-1)^2+y^2} = 4 - \sqrt{(x+1)^2+y^2}$$

$$x^2-2x+1+y^2 = 16 - 8\sqrt{(x+1)^2+y^2} + x^2+2x+1+y^2$$

$$-4x-16 = -8\sqrt{(x+1)^2+y^2}$$

$$x+4 = 2\sqrt{(x+1)^2+y^2}$$

$$\frac{x}{2} + 2 = \sqrt{(x+1)^2+y^2}$$

$$\frac{x^2}{4} + 2x + 4 = x^2 + 2x + 1 + y^2$$

$$\frac{3}{4}x^2 + y^2 = 3$$

$$\frac{x^2}{4} + \frac{y^2}{3} = 1$$

Q13.

EXT. 2

$$(a) \quad x^3 + 2x^2 - 15x - 36 = (x+p)^2(x+q)$$

"-p" is a double root

$$f'(x) = 3x^2 + 4x - 15$$

$$\text{Let } f'(x) = 0 \Rightarrow 3x^2 + 4x - 15 = 0$$

$$(3x-5)(x+3) = 0$$

$$\begin{array}{r} 3x \quad -5 \\ \times \quad 3 \\ \hline x \end{array}$$

$$x = \frac{5}{3} \text{ or } x = -3.$$

$$\text{Then } f(-3) = -27 + 18 + 45 - 36 = 0$$

$$\therefore \underline{p=3}$$

$$\Rightarrow f(x) = (x+3)^2(x+q)$$

$$\Rightarrow x^3 + 2x^2 - 15x - 36 = (x^2 + 6x + 9)(x+q)$$

$$\text{By inspection, } \underline{q = -4.}$$

$$\therefore \underline{p=3}, \underline{q=-4}$$

$$(b) \quad \frac{x^2}{9} + \frac{y^2}{4} = 1$$

$$(i) \quad e = \frac{\sqrt{a^2 - b^2}}{a}$$

$$= \frac{\sqrt{9-4}}{3}$$

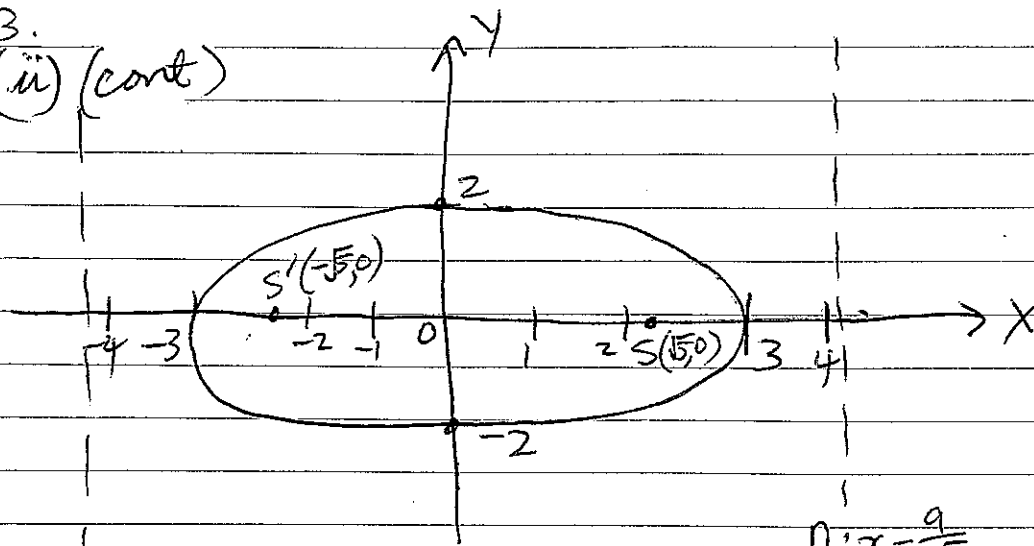
$$= \frac{\sqrt{5}}{3}$$

$$(ii) \quad S = (\pm ae, 0) = (\pm \sqrt{5}, 0)$$

$$D: x = \pm \frac{a}{e} \Rightarrow x = \pm \frac{3}{(\sqrt{5}/3)}$$

$$\underline{x = \pm \frac{9}{\sqrt{5}}}$$

13.  
(b)(ii) (cont)



$$D': x = -\frac{9}{\sqrt{5}}$$

$$D: x = \frac{9}{\sqrt{5}}$$

2

(iii)  $\frac{x^2}{9} + \frac{y^2}{4} = 1$

$$\Rightarrow \frac{2x}{9} + \frac{y}{2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{4x}{9y}$$

At  $P(3\cos\theta, 2\sin\theta)$ ,  $\frac{dy}{dx} = \frac{-12\cos\theta}{18\sin\theta}$

$$\frac{dy}{dx} = -\frac{2\cos\theta}{3\sin\theta}$$

Then Eqn of tangent:  $y - y_1 = m(x - x_1)$

$$y - 2\sin\theta = \frac{-2\cos\theta}{3\sin\theta} (x - 3\cos\theta)$$

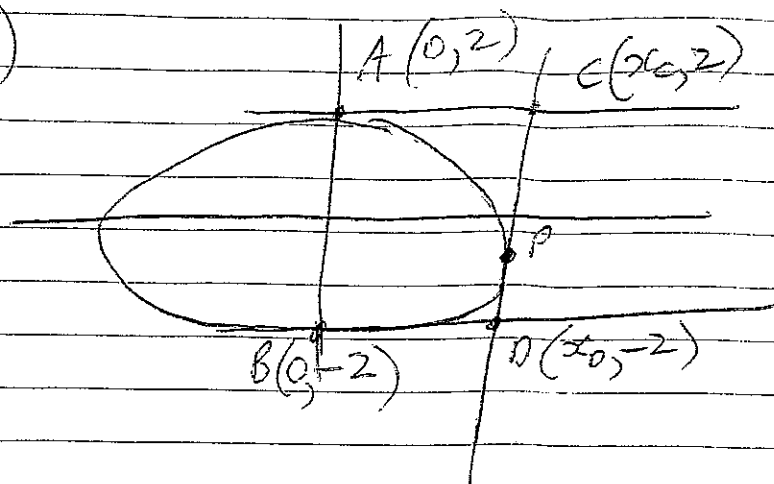
$$3y\sin\theta - 6\sin^2\theta = -2x\cos\theta + 6\cos^2\theta$$

$$\Rightarrow 2x\cos\theta + 3y\sin\theta = 6(\sin^2\theta + \cos^2\theta)$$

$$2x\cos\theta + 3y\sin\theta = 6 \quad \#$$

3

13. (iv)



Eqn of tangent at P :  $2x \cos \theta + 3y \sin \theta = 6$  (1)

Eqn tangent at A :  $y = 2$  (2)

Eqn tangent at B :  $y = -2$  (3)

For C Sub (2) in (1)  $\Rightarrow 2x \cos \theta + 6 \sin \theta = 6$   $\therefore$   
 $2x \cos \theta = 6(1 - \sin \theta)$   
 $x = \frac{6(1 - \sin \theta)}{2 \cos \theta}$

$x_c = \frac{3(1 - \sin \theta)}{\cos \theta}$

$\therefore C = \left( \frac{3(1 - \sin \theta)}{\cos \theta}, 2 \right)$  ✓

For D Sub (3) in (1)  $\Rightarrow 2x \cos \theta - 6 \sin \theta = 6$   
 $2x \cos \theta = 6(1 + \sin \theta)$

$x_0 = \frac{3(1 + \sin \theta)}{\cos \theta}$

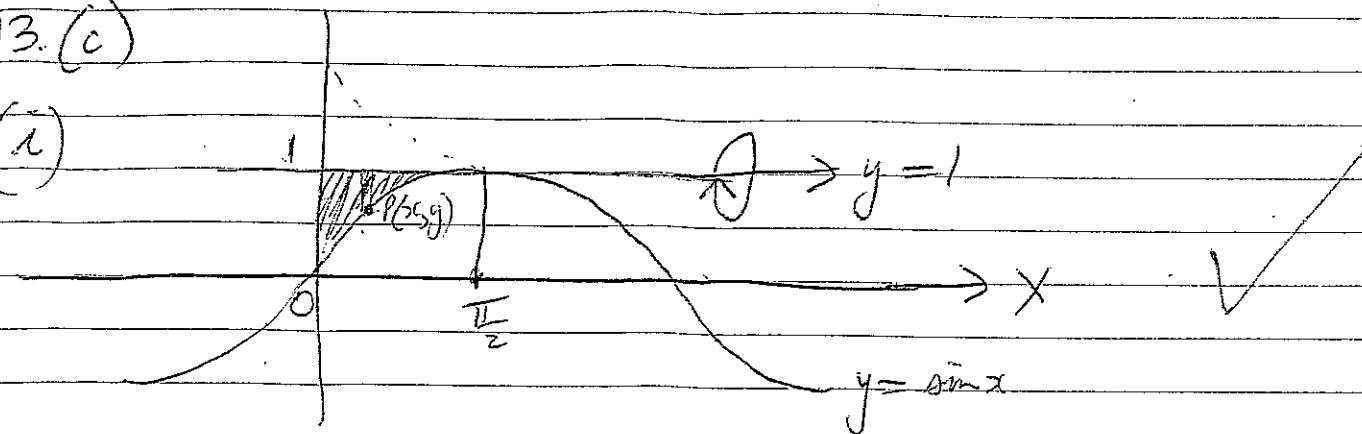
$\therefore D = \left( \frac{3(1 + \sin \theta)}{\cos \theta}, -2 \right)$  ✓ (3)

Then  $AC = \frac{3(1 - \sin \theta)}{\cos \theta}$  and  $BD = \frac{3(1 + \sin \theta)}{\cos \theta}$

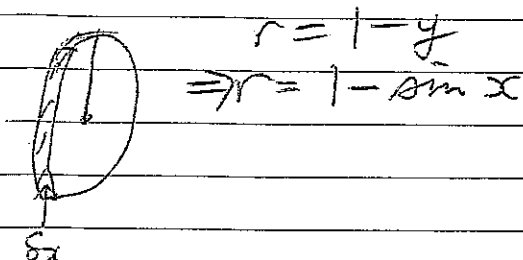
$\therefore AC \times BD = \frac{3(1 - \sin \theta)}{\cos \theta} \times \frac{3(1 + \sin \theta)}{\cos \theta} = \frac{9(1 - \sin^2 \theta)}{\cos^2 \theta}$   
 $= 9$  ✓

13. (c)

(i)



(ii) Washers.



$$V_{\text{slice}} = \pi r^2 \delta x$$

$$= \pi (1 - \sin x)^2 \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\frac{\pi}{2}} \pi (1 - \sin x)^2 \delta x$$

$$V = \int_0^{\frac{\pi}{2}} \pi (1 - \sin x)^2 dx$$

$$= \pi \int_0^{\frac{\pi}{2}} (1 - 2\sin x + \sin^2 x) dx$$

$$= \pi \int_0^{\frac{\pi}{2}} (1 - 2\sin x + \frac{1}{2} - \frac{1}{2}\cos 2x) dx$$

$$= \pi \int_0^{\frac{\pi}{2}} (\frac{3}{2} - 2\sin x - \frac{1}{2}\cos 2x) dx$$

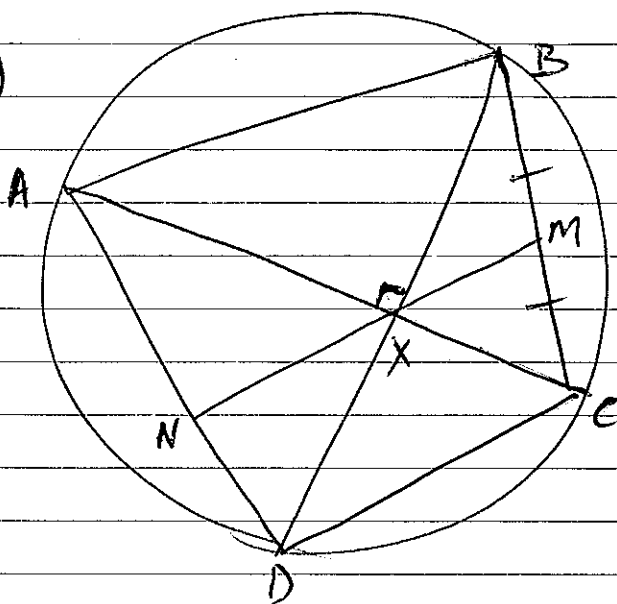
$$= \pi \left[ \frac{3}{2}x + 2\cos x - \frac{1}{4}\sin 2x \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left[ \left( \frac{3\pi}{4} + 0 - 0 \right) - (0 + 2 - 0) \right]$$

$$= \left( \frac{3\pi^2}{4} - 2\pi \right) \text{ units}^3$$

4

14) a) i)



Since  $\angle BXC = 90^\circ$

BC is the diameter of circle BCX

Since M is the midpoint of BC

M is the centre of circle BCX

$BM = MX$  (equal radii)

ii)  $\triangle MBX$  is isosceles ( $BM = MX$ )

$\therefore \angle MBX = \angle MXB$  (base angles of isosceles triangle)

iii) Let  $\angle MBX = \angle MXB = \alpha$

$\angle BCX = 90 - \alpha$  (angle sum of  $\triangle BCX$ )

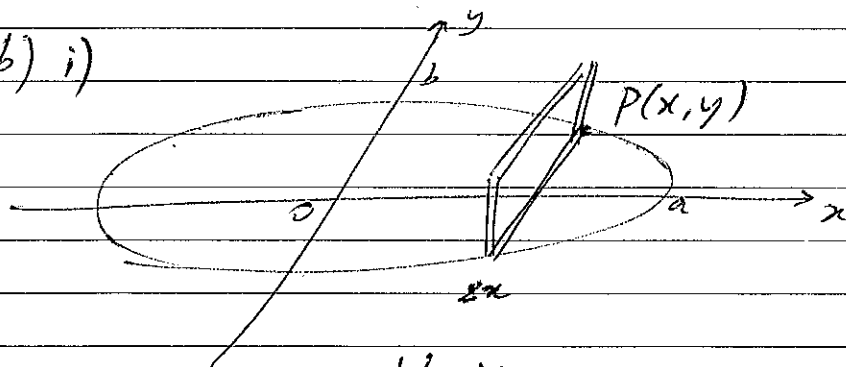
$\angle BDN = 90 - \alpha$  (angles in the same segment)

$\angle NXD = \alpha$  (vertically opposite angles)

$\angle XND = 90^\circ$  (angle sum of  $\triangle NXD$ )

$\therefore MN \perp AD$

b) i)



$$\begin{aligned} \Delta V &= bh \Delta x \\ \Delta V &= (2y)(y) \Delta x \\ &= 2y^2 \Delta x \\ &= \frac{2b^2(a^2 - x^2)}{a^2} \Delta x \end{aligned}$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{y^2}{b^2} = 1 - \frac{x^2}{a^2}$$

$$y^2 = b^2 \left( 1 - \frac{x^2}{a^2} \right)$$

$$y^2 = \frac{b^2}{a^2} (a^2 - x^2)$$

$$ii) V = \lim_{\Delta x \rightarrow 0} \sum_{x=-a}^a \frac{2b^2}{a^2} (a^2 - x^2) \Delta x$$

$$= \frac{2b^2}{a^2} \int_{-a}^a (a^2 - x^2) dx$$

$$= \frac{4b^2}{a^2} \int_0^a (a^2 - x^2) dx$$

$$= \frac{4b^2}{a^2} \left[ a^2 x - \frac{x^3}{3} \right]_0^a$$

$$= \frac{4b^2}{a^2} \left[ a^2(a) - \frac{(a)^3}{3} - (0) \right]$$

$$= \frac{4b^2}{a^2} \left[ \frac{2a^3}{3} \right]$$

$$= \frac{8ab^2}{3} \text{ cubic units.}$$

$$c) i) (x+y)^2 = (x-y)^2 + 4xy$$

$$> 4xy$$

$$ii) \text{ similarly } (z+w)^2 > 4zw$$

$$(x+y)^2 (z+w)^2 > 4xy \cdot 4zw$$

$$= 16xyzw$$

$$\therefore [(x+y)(z+w)]^2 > 16xyzw$$

$$iii) \text{ From (i) } \left( \frac{x+y}{4} + \frac{z+w}{4} \right)^2 > 4 \left( \frac{x+y}{4} \right) \left( \frac{z+w}{4} \right)$$

$$\left( \frac{x+y+z+w}{4} \right)^2 > \frac{(x+y)(z+w)}{4}$$

$$\left( \frac{x+y+z+w}{4} \right)^4 > \frac{[(x+y)(z+w)]^2}{16}$$

From (ii)

$$\left( \frac{x+y+z+w}{4} \right)^4 \geq \frac{16xyzw}{16}$$

$$\frac{x+y+z+w}{4} \geq \sqrt[4]{xyzw}$$

iv) let  $x = \frac{a}{b}, y = \frac{b}{c}, z = \frac{c}{d}, w = \frac{d}{a}$

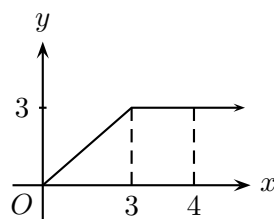
$$\frac{\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}}{4} \geq \sqrt[4]{\frac{a}{b} \cdot \frac{b}{c} \cdot \frac{c}{d} \cdot \frac{d}{a}}$$

$$\frac{\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}}{4} \geq 1$$

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4$$

2014 Extension 2 Mathematics Trial HSC:  
Solutions— Question 15

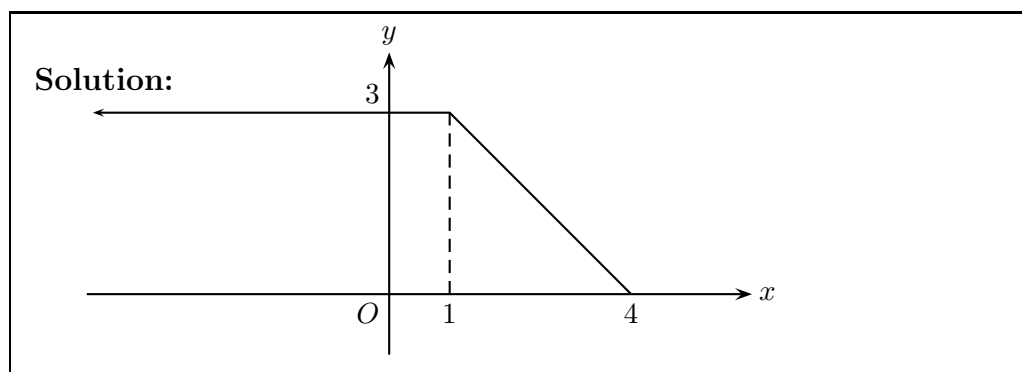
15. (a) The graph of  $y = f(x)$  is shown.



Sketch the following on separate diagrams:

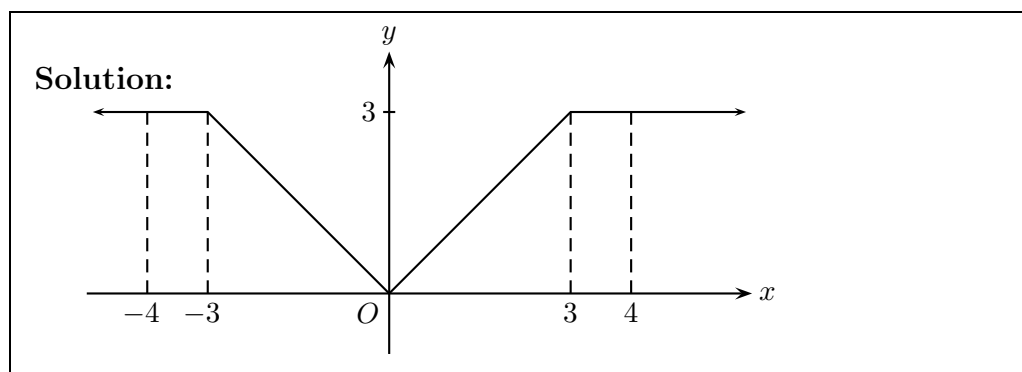
- (i)  $y = f(4 - x)$ ,

1



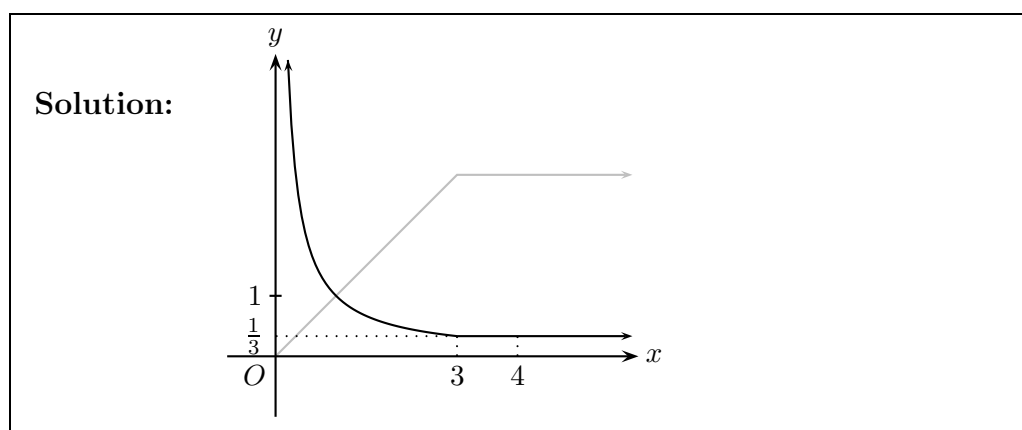
- (ii)  $y = f(|x|)$ ,

1



- (iii)  $y \times f(x) = 1$ ,

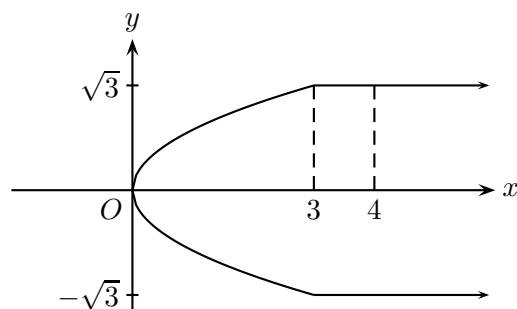
1



(iv)  $y^2 = f(x)$ .

1

**Solution:**



(b) Let  $w$  be a non-real cube root of unity.

(i) Show that  $1 + w + w^2 = 0$ .

1

**Solution:**

$$\begin{aligned} w^3 &= 1, \\ w^3 - 1 &= 0, \\ (w - 1)(w^2 + w + 1) &= 0, \\ \text{but } w &\neq 1 \text{ as } w \text{ not real,} \\ \therefore w^2 + w + 1 &= 0. \end{aligned}$$

(ii) Simplify  $(1 + w)^2$ .

1

**Solution:**

$$\begin{aligned} (1 + w)^2 &= w^2 + 2w + 1, \\ &= (w^2 + w + 1) + w, \\ &= w. \end{aligned}$$

(iii) Show that  $(1 + w)^3 = -1$ .

1

**Solution:**

$$\begin{aligned} (1 + w)^2(1 + w) &= w(1 + w), \\ &= w + w^2, \\ &= (1 + w + w^2) - 1, \\ &= -1. \end{aligned}$$

(iv) Using part (iii) simplify  $(1 + w)^{3n}$  where  $n \in \mathbb{Z}^+$ .

1

**Solution:**

$$\begin{aligned} ((1 + w)^3)^n &= (-1)^n, \\ &= \begin{cases} -1 & \text{if } n \text{ is odd,} \\ 1 & \text{if } n \text{ is even.} \end{cases} \end{aligned}$$

(v) Show that

3

$$\begin{aligned} \binom{3n}{0} - \frac{1}{2} \left[ \binom{3n}{1} + \binom{3n}{2} \right] + \binom{3n}{3} - \frac{1}{2} \left[ \binom{3n}{4} + \binom{3n}{5} \right] + \binom{3n}{6} - \dots \\ \dots + \binom{3n}{3n} = (-1)^n \end{aligned}$$

[Hint: You may use  $\Re(w) = \Re(w^2) = -\frac{1}{2}$ ].

**Solution:** Now from part (iv),  $(1+w)^{3n} = (-1)^n \in \mathbb{R}$ ,  
so when looking at the expansion of  $(1+w)^{3n}$   
we need only consider the real parts.  
We also note that  $w^{3k} = 1$  as  $w^3 = 1$ ,  $w^{3k+1} = w$ ,  
 $w^{3k+2} = w^2$  and,  
using  $\Re(w) = \Re(w^2) = -\frac{1}{2}$ , we have  

$$(1+w)^{3n} = \binom{3n}{0} + \binom{3n}{1}w + \binom{3n}{2}w^2 + \binom{3n}{3}w^3 + \binom{3n}{4}w^4 + \binom{3n}{5}w^5 + \dots$$

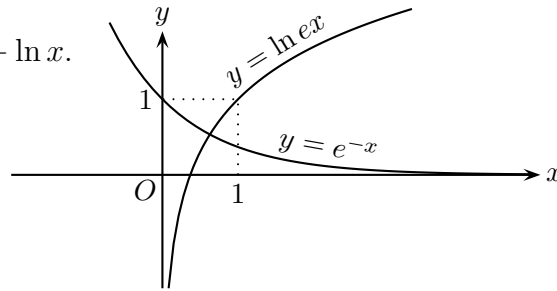
$$\dots + \binom{3n}{3n-2}w^{3n-2} + \binom{3n}{3n-1}w^{3n-1} + \binom{3n}{3n}w^{3n},$$
*i.e.*  $(-1)^n = \binom{3n}{0} - \frac{1}{2}[\binom{3n}{1} + \binom{3n}{2}] + \binom{3n}{3} - \frac{1}{2}[\binom{3n}{4} + \binom{3n}{5}] + \binom{3n}{6} - \dots$ 

$$\dots + \binom{3n}{3n}.$$

- (c) (i) Show that  $\ln(ex) > e^{-x}$  for  $x \geq 1$ . (Use a diagram.)

1

**Solution:**  $\ln ex = 1 + \ln x$ .



- (ii) Hence show that  $\ln(e^n \times n!) > \frac{e^n - 1}{e^n(e - 1)}$ .

3

**Solution: Method 1—**

From part (i),  $\ln(ex) > e^{-x}$ ;

$$\begin{aligned} \text{so L.H.S.} &= \ln(1 \times e) + \ln(2e) + \ln(3e) + \dots + \ln((n-1)e) + \ln(ne), \\ &> e^{-1} + e^{-2} + e^{-3} + \dots + e^{1-n} + e^{-n}, \\ &> \frac{1}{e^n}(e^{n-1} + e^{n-2} + \dots + e^2 + e^1 + e^0), \\ &> \frac{1}{e^n} \times \frac{e^n - 1}{e - 1}. \\ \text{i.e. } \ln(e^n \times n!) &> \frac{e^n - 1}{e^n(e - 1)}. \end{aligned}$$

**Solution: Method 2—**

Test  $n = 1$ ,

$$\begin{aligned} \text{L.H.S.} &= \ln e, & \text{R.H.S.} &= \frac{e - 1}{e(e - 1)}, \\ &= 1. & &= \frac{1}{e}. \end{aligned}$$

So it is true for  $n = 1$ .

Now assume true for some  $n = k$ ,  $k \in \mathbb{Z}^+$ ,

$$\text{i.e. } \ln(e^k \times k!) > \frac{e^k - 1}{e^k(e - 1)}.$$

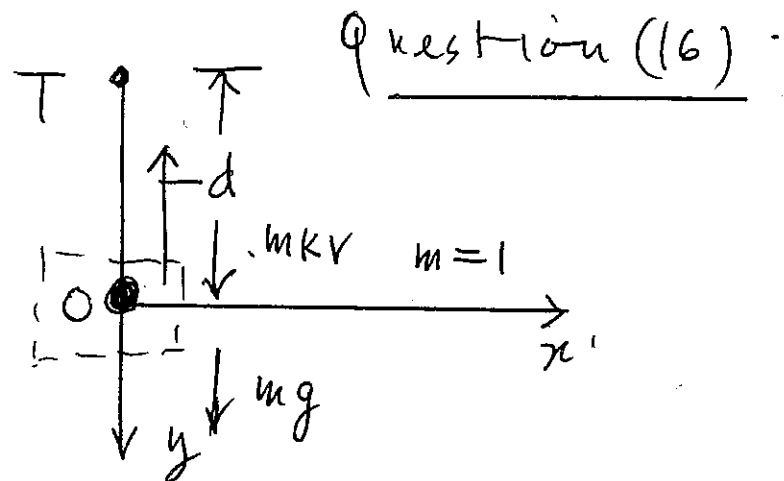
Then test for  $n = k + 1$ , i.e.  $\ln(e^{k+1} \times (k+1)!) > \frac{e^{k+1} - 1}{e^{k+1}(e-1)}$ .

$$\begin{aligned}\text{L.H.S.} &= \ln(e^k \cdot k! \times e(k+1)), \\ &= \ln(e^k \times k!) + \ln(e(k+1)).\end{aligned}$$

Now  $\ln(e^k \times k!) > \frac{e^k - 1}{e^k(e-1)}$  from the assumption,  
and  $\ln(e(k+1)) > e^{-(k+1)}$  from part (i) where  $x \geq 1$ ,

$$\begin{aligned}\therefore \text{L.H.S.} &> \frac{e^k - 1}{e^k(e-1)} \times \frac{e}{e} + \frac{1}{e^{k+1}} \times \frac{e-1}{e-1}, \\ &> \frac{e^{k+1} - e + e - 1}{e^{k+1}(e-1)}, \\ &> \frac{e^{k+1} - 1}{e^{k+1}(e-1)} = \text{R.H.S.}\end{aligned}$$

Thus true for  $n = k + 1$  if true for  $n = k$ , but true for  $n = 1$  so true for  $n = 2, 3, 4$ , and so on for all  $n \in \mathbb{Z}^+$ .



$$(i) \quad \ddot{x} = g - kv \quad [1]$$

$$(ii) \quad \frac{dv}{dt} = g - kv.$$

$$\frac{dt}{dv} = \frac{1}{g - kv}.$$

$$\therefore t = -\frac{1}{k} \int \left( \frac{-k}{g - kv} \right) dv.$$

$$t = -\frac{1}{k} \ln(g - kv) + c_1$$

When  $t=0$ ,  $v=0$ .

$$\therefore 0 = -\frac{1}{k} \ln(g - kU) + c_1$$

$$\therefore c_1 = \frac{1}{k} \ln(g - kU).$$

i.e.  $t = \frac{1}{k} \ln \left( \frac{g - kU}{g - kv} \right), \quad [2]$

$$\therefore e^{-kt} = \frac{g - kv}{g - kU}.$$

$$\therefore g - kv = (g - kU) e^{-kt}$$

$$\therefore kv = g - (g - kU) e^{-kt}$$

$$v_p = \frac{g}{k} - \frac{g - kU}{k} e^{-kt} \quad [1] \quad \text{--- (1)}$$

$$(iii) \quad \frac{dx}{dt} = \frac{g}{k} - \left( \frac{g - kU}{k} \right) e^{-kt}$$

$$x = \frac{gt}{k} + \left( \frac{g - kU}{k^2} \right) \int (-k) e^{-kt} dt$$

$$x = \frac{g}{k} t + \left( \frac{g - kU}{k^2} \right) e^{-kt} + c_2.$$

When  $t=0$ ,  $x=-d$ .

$$\therefore -d = \frac{g - kU}{k^2} + c_2 \quad [2]$$

$$\Rightarrow c_2 = -\left( \frac{g - kU}{k^2} \right) - d.$$

$$\therefore x_p = \left( \frac{gt - kd}{k} \right) + \frac{g - kU}{k^2} (e^{-kt} - 1)$$

$$\text{--- (2)}$$

(iv) ——— ①

$$v_p = \frac{g}{k} - \frac{g - kU}{k} e^{-kt}$$

$$x_p = \left( \frac{gt - kd}{k} \right) + \frac{g - kU}{k^2} (e^{-kt} - 1)$$

————— ②

put  $U = 0$

$$\therefore v_p = \frac{g}{k} (1 - e^{-kt}) \quad [1]$$

————— ③

put  $U = 0, d = 0$

$$\Rightarrow x_p = \frac{gt}{k} + \frac{g}{k^2} (e^{-kt} - 1)$$

————— ④ [1]

(v) The particles collide

when  $x_p = x_q$

$$\cancel{\frac{gt - kd}{k}} + \left( \cancel{\frac{g - kU}{k^2}} \right) (e^{-kt} - 1) = \frac{gt}{k} + \frac{g}{k^2} (e^{-kt} - 1)$$

$$\text{i.e. } \frac{U}{k} e^{-kt} = \frac{U}{k} - d$$

$$e^{-kt} = 1 - \frac{kd}{U}$$

$$-kt = \ln \left( 1 - \frac{kd}{U} \right)$$

$$t = -\frac{1}{k} \ln \left( 1 - \frac{kd}{U} \right) \quad [1]$$

————— ⑤

When  $t = -\frac{1}{k} \ln \left( \frac{U}{U - kd} \right)$

$$v_p = \frac{g}{k} - \frac{g}{k} + U + (g - kU) \frac{d}{U}$$

$$= U + \frac{gd}{U} - kd \quad \text{————— 6a}$$

$$v_q = \frac{g}{k} \left( 1 - \left( 1 + \frac{kd}{U} \right) \right) = -\frac{gd}{U} \quad [2]$$

————— ⑥

$$\therefore \text{Speed of Collision } |v_p - v_q|$$

$$= \left| \left( U + \frac{gd}{U} - kd \right) - \left( -\frac{gd}{U} \right) \right|$$

$$= |U - kd| \quad \text{————— ⑦}$$

### Question 16 (b)

$$\begin{aligned}
 \text{(i)} \quad & \sin(2r+1)\theta - \sin(2r-1)\theta \\
 &= \sin 2r\theta \cos \theta + \cos 2r\theta \sin \theta \\
 &\quad - (\sin 2r\theta \cos \theta - \cos 2r\theta \sin \theta) \\
 &= 2 \sin \theta \cos 2r\theta. \quad [1]
 \end{aligned}$$

OR.

$$\boxed{\sin A - \sin B = 2 \cos \left( \frac{A+B}{2} \right) \sin \left( \frac{A-B}{2} \right)}$$

$$\begin{aligned}
 & \sin(2r+1)\theta - \sin(2r-1)\theta \\
 &= 2 \cos \left[ \frac{(2r+1)\theta + (2r-1)\theta}{2} \right] \sin \left[ \frac{(2r+1)\theta - (2r-1)\theta}{2} \right] \\
 &= 2 \cos \left( \frac{4r\theta}{2} \right) \sin \left( \frac{2\theta}{2} \right) \\
 &= 2 \cos(2r\theta) \sin \theta. \quad \text{--- (1)}
 \end{aligned}$$

(ii) From (i) [1]

$$2 \sin \theta \sum_{r=1}^n \cos(2r\theta) = \sum_{r=1}^n [\sin(2r+1)\theta - \sin(2r-1)\theta]$$

$$\therefore \sin \theta \sum_{r=1}^n \cos(2r\theta) = \frac{1}{2} \sum_{r=1}^n [\sin(2r+1)\theta - \sin(2r-1)\theta]$$

$$\begin{aligned}
 &= \frac{1}{2} [(\cancel{\sin 3\theta} - \sin \theta) + (\sin 5\theta - \cancel{\sin 3\theta}) + \dots + \sin(2n+1)\theta - \cancel{\sin(2n-1)\theta}] \\
 &= \frac{1}{2} [\sin(2n+1)\theta - \sin \theta]. \quad [1]
 \end{aligned}$$

$$\text{(iii)} \quad \boxed{\cos^2 r \left( \frac{\pi}{100} \right) = \frac{1 + \cos \left[ 2r \left( \frac{\pi}{100} \right) \right]}{2}}$$

$$\therefore \cos 2r \left( \frac{\pi}{100} \right) = \frac{1}{2} + \frac{1}{2} \cos \left( \frac{2r\pi}{100} \right)$$

$$\therefore \sum_{r=1}^n \cos 2r\theta = \frac{1}{2 \sin \theta} [\sin(2n+1)\theta - \sin \theta] \quad \text{--- (2)}$$

$$\begin{aligned}
 \therefore \sum_{r=1}^{100} \cos^2 \left( \frac{r\pi}{100} \right) &= \sum_{r=1}^{100} \left[ \frac{1}{2} + \frac{1}{2} \cos \left( 2r \left( \frac{\pi}{100} \right) \right) \right] \\
 &= \frac{1}{2} \times 100 + \frac{1}{2} \sum_{r=1}^{100} \cos \left( 2r \left( \frac{\pi}{100} \right) \right) \quad [1]
 \end{aligned}$$

Using (2)  $\theta = \frac{\pi}{100}, n = 100$

$$\begin{aligned}
 \therefore \sum_{r=1}^{100} \cos^2 \left( \frac{r\pi}{100} \right) &= 50 + \frac{1}{2} \left[ \frac{\sin \left( \frac{201\pi}{100} \right) - \sin \frac{\pi}{100}}{2 \sin \left( \frac{\pi}{100} \right)} \right] \\
 &= 50 + \frac{\sin \left( 2\pi + \frac{\pi}{100} \right) - \sin \left( \frac{\pi}{100} \right)}{4 \sin \left( \frac{\pi}{100} \right)} \quad [1] \\
 &= 50 + \frac{(\sin \frac{\pi}{100} - \sin \frac{\pi}{100})}{4 \sin \frac{\pi}{100}} \quad \left[ \begin{array}{l} \sin(\theta + 2\pi) \\ = \sin \theta \end{array} \right] \\
 &= 50
 \end{aligned}$$